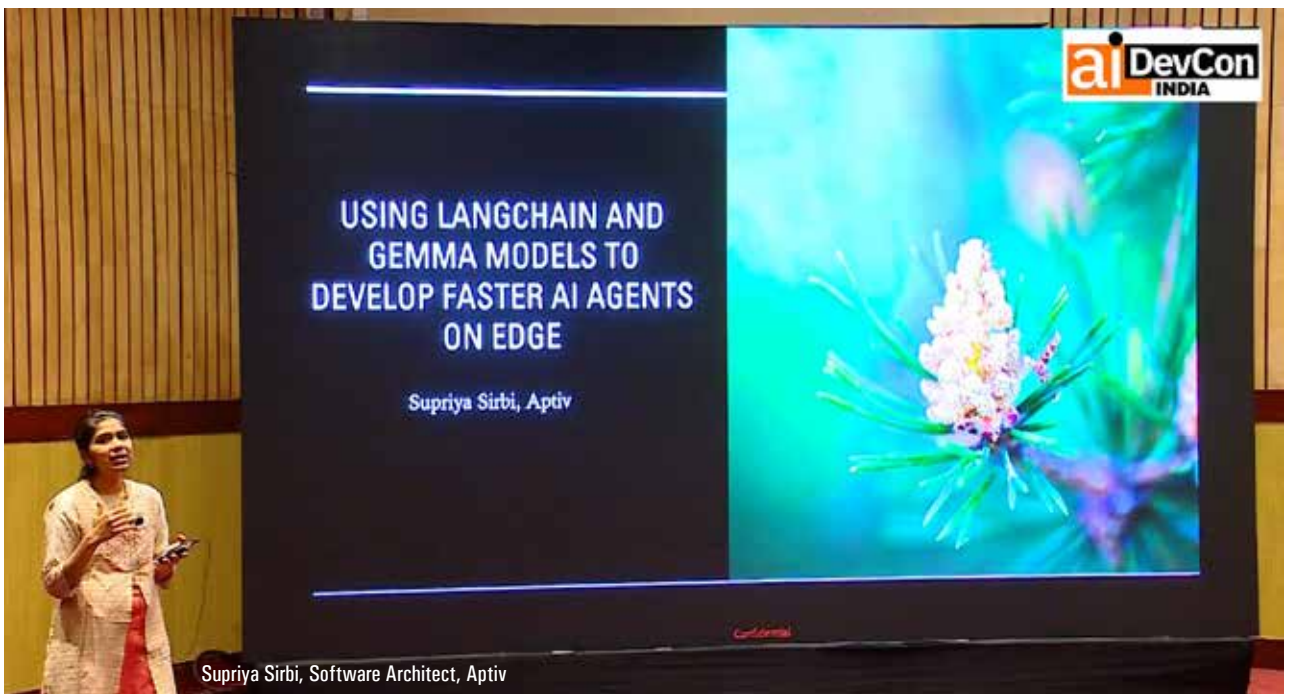
Building Lightweight AI Agents for Edge Computing

Lightweight AI models are making it possible to run intelligent applications directly on local devices. LangChain, Gemma and PaliGemma simplify AI agent development for edge computing while reducing reliance on cloud infrastructure.



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Artificial intelligence (AI) is increasingly moving beyond cloud-based environments and onto local devices. The change has created a growing need for lightweight language models that can intelligently respond without constant access to the internet. Running AI at the edge enables applications to operate in remote locations, improves privacy and reduces dependence on cloud infrastructure, making it suitable for many real-world scenarios.

One of the key enablers of this transition is Google's Gemma family of open source models. Gemma models have been designed with smaller parameter sizes, and hence, they can be run on devices that have relatively modest hardware capabilities but can support various AI applications. In combination with other frameworks such as LangChain, developers can build intelligent agents that can perform tasks like image captioning, summarisation, and reasoning using only local systems.

Why lightweight models matter for edge AI

Running an AI model on edge devices has its own limitations and trade-offs must be made between inference performance and hardware limitations. While large language models use billions of parameters, sometimes even hundreds of billions, Gemma models have been designed in a way that they can function using merely 2 to 3 billion parameters, which is enough to function with 16GB RAM or even lower.

Edge AI becomes significant when there is no internet connection available. Regardless of whether they are working in remote areas such as forests or mountains where no internet is available, the models continue processing data locally and generate responses without internet connectivity. Since edge devices have limited computing resources, lightweight and fine-tuned models are better suited for these specialised use cases than large, general-purpose models.

Although lightweight models may introduce slightly higher inference times compared with powerful cloud-hosted